

Poisson's equation

Task 9

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MA18C014

$$\frac{\partial^2 u}{\partial x^2} + f = 0$$

$u(a) = u(b) = 0 \leftarrow$ Dirichlet boundary conditions

Weak formulation

To find the weak formulation, we multiply by an appropriate test function v and integrate over the domain

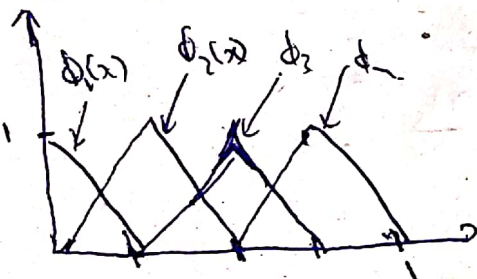
$$\frac{\partial^2 u}{\partial x^2} + f = 0 \Rightarrow \int_a^b v \left(\frac{\partial^2 u}{\partial x^2} + f \right) dx = 0 \quad \forall v \in \text{Test functions}$$

$$\Rightarrow \left[- \int_a^b \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx + \int_a^b v f dx \right] = 0 \quad \text{Weak formulation}$$

Let $\phi_j(x) : j=1, \dots, n$ form a basis. $u(x) \approx \sum u_j \phi_j(x)$

where $\phi_j(x) : j=1, \dots, n$ form a basis.

Basis functions: choose $a=0$ $b=1$



$$\Rightarrow - \sum u_j \int_a^b \frac{\partial \phi_j}{\partial x} \frac{\partial \phi_i}{\partial x} dx + \int_a^b f \phi_i dx = 0 \quad \forall \phi_i \in \text{Test functions } i=1, \dots, n$$

$$\therefore \sum u_j \left(\int_a^b \frac{\partial \phi_j}{\partial x} \frac{\partial \phi_i}{\partial x} dx \right) + \int_a^b f \phi_i dx = 0 \quad \forall \phi_i \in \text{Basis } i$$

$$A_{ij} = \frac{1}{x_j - x_{j-1}} + \frac{1}{x_{j+1} - x_j}$$

$$b_j = \frac{(x_{j+1} - x_{j-1})}{2} f(x_j)$$

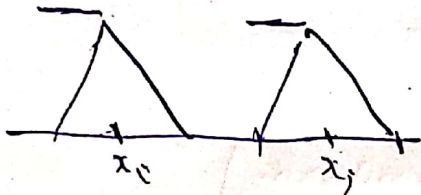
We solve $Au=b$ in MATLAB. The solution gives us the coefficients u_j in $u \approx \sum u_j \phi_j$
 $\therefore u(x)$ can be reconstructed

Solving Poisson's equation using the piecewise linear basis

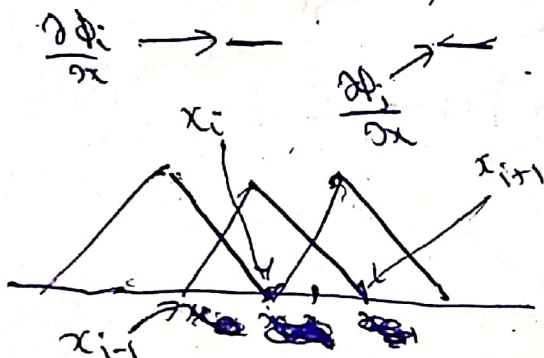
We have $A_{ij} = - \int_0^1 \frac{\partial \phi_j}{\partial x} \frac{\partial \phi_i}{\partial x} dx$ for $j = 1, 2, \dots, n-1$
 ($\because u(0) = u(1) = 0$, $j = 0, j = n$ don't matter)

$$b_j = \int_0^1 \phi_j f dx$$

$\{A_{ij}\}$ is sparse because if i, j are not neighbours, then $A_{ij} = 0$



$\leftarrow$ No overlap $\Rightarrow \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} = 0$
 $\Rightarrow A_{ij} = 0$



$$\therefore A_{ij} = \begin{cases} 0 & |i-j| > 1 \\ \text{nonzero} & |i-j| \leq 1 \end{cases}$$

$$A_{ii} = - \int_0^1 \left(\frac{\partial \phi_i}{\partial x} \right)^2 dx = - \int_{x_{i-1}}^{x_i} \left(\frac{\partial \phi_i}{\partial x} \right)^2 dx + \int_{x_i}^{x_{i+1}} \left(\frac{\partial \phi_i}{\partial x} \right)^2 dx$$

$$= - \left(\frac{1}{x_i - x_{i-1}} - \frac{1}{x_{i+1} - x_i} \right)$$

$$A_{i,i+1} = - \int_0^1 \left(\frac{\partial \phi_i}{\partial x} \right) \left(\frac{\partial \phi_{i+1}}{\partial x} \right) dx$$

$$= - \int_{x_i}^{x_{i+1}} \left(\frac{\partial \phi_i}{\partial x} \right) \left(\frac{\partial \phi_{i+1}}{\partial x} \right) dx$$

$$= - \frac{1}{x_{i+1} - x_i} \cdot \frac{1}{x_{i+1} - x_i}$$

Actual solution of $u'' + 2 = 0$ $u(0) = 0 = u(1)$

$$\frac{d^2 u}{dx^2} + 2 = 0$$

Here $\frac{d}{dx} = \frac{d}{dx}$

$$\frac{d^2 u}{dx^2} + 2 = 0 \Rightarrow \frac{d^2 u}{dx^2} = -2$$

$$\Rightarrow \frac{du}{dx} = -2x + C_1$$

$$\Rightarrow \boxed{u = -x^2 + C_1 x + C_2}$$

$$u(0) = 0 \Rightarrow 0 = C_2$$

$$u(1) = 0 \Rightarrow 0 = -1 + C_1 \Rightarrow C_1 = 1$$

$\therefore$ The actual solution is $u(x) = -x^2 + x$

From the matlab code we observe that the numerical solution matches with the actual solution at all grid points.